Building Materials

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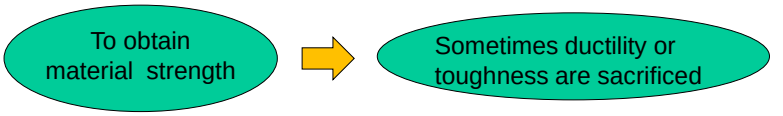
Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
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1. Introduction
2. Crystalline structure
3. Production and Alloys
4. Mechanical properties
5. **Strengthening mechanisms**
6. **Corrosion**
7. **Engineering application**

3.5 Strengthening Mechanism

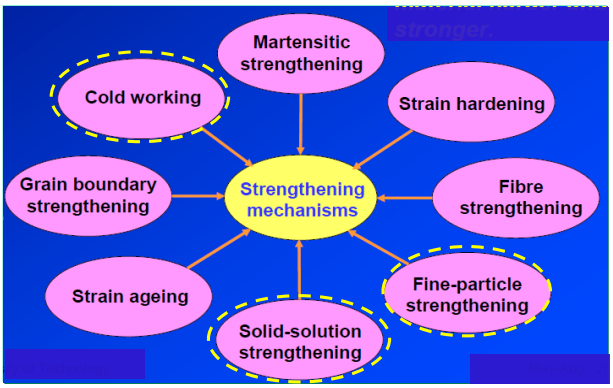
Why strengthening mechanisms?

We can tailor the mechanical properties of an engineering material by choosing between strength and ductility.



Strengthening techniques rely on restricting dislocation motion to render a material harder and stronger.

3.5 Strengthening Mechanism



They make dislocation motion more difficult!



----- These mechanisms may be combined.
----- For example, the world's strongest structural material (with some ductility) is steel piano wire. It combines four strengthening mechanisms, and can have a yield strength of 3500MPa. One wire, 2mm in diameter, can hold up a 2,000kg Ford Explorer.

3.5 Strengthening Mechanism

Cold Working

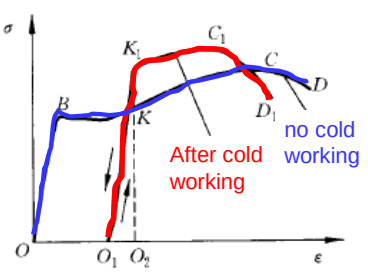
☐ Ductile material becomes harder and stronger as it is plastically deformed

strength ↑
ductile ↓

Percent of cold work:

$$\%CW = \left(\frac{A_0 - A_d}{A_0} \right) \times 100$$

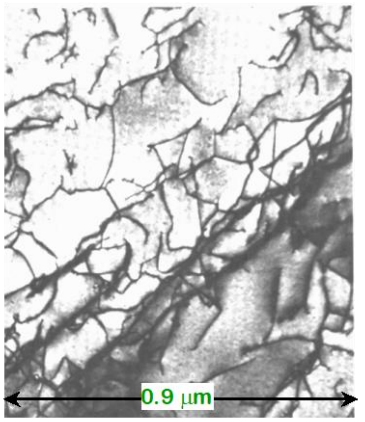
%CW: percent of cold work
A0: original cross-sectional area
Ad: area after deformation



3.5 Strengthening Mechanism

Cold Working

Ti alloy after cold working:

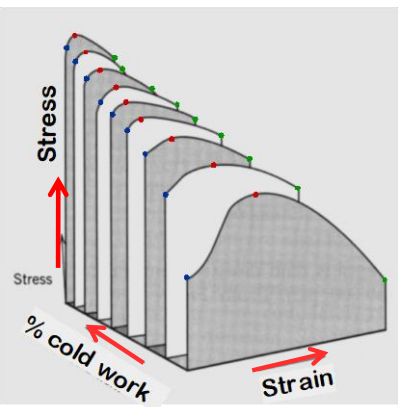


- Dislocations entangle with one another during cold work.
- Further dislocation motion Becomes more difficult.

3.5 Strengthening Mechanism

Cold Working

Impact of Cold Work

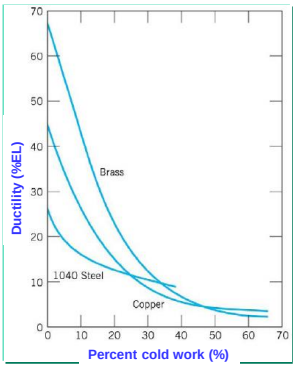


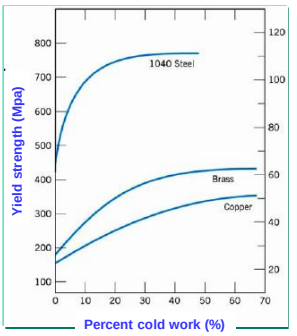
- Yield strength increases.
- Tensile strength increases.
- Ductility decreases.

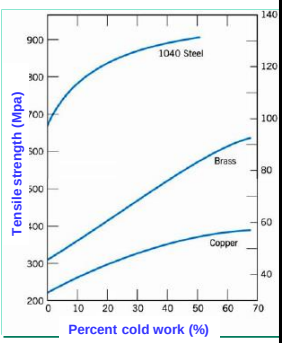
3.5 Strengthening Mechanism

Cold Working

As the yield strength and the tensile strength increases with increasing amount of cold working, the ductility of the metal decreases.



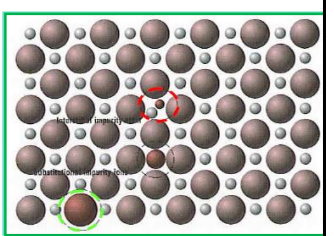




3.5 Strengthening Mechanism

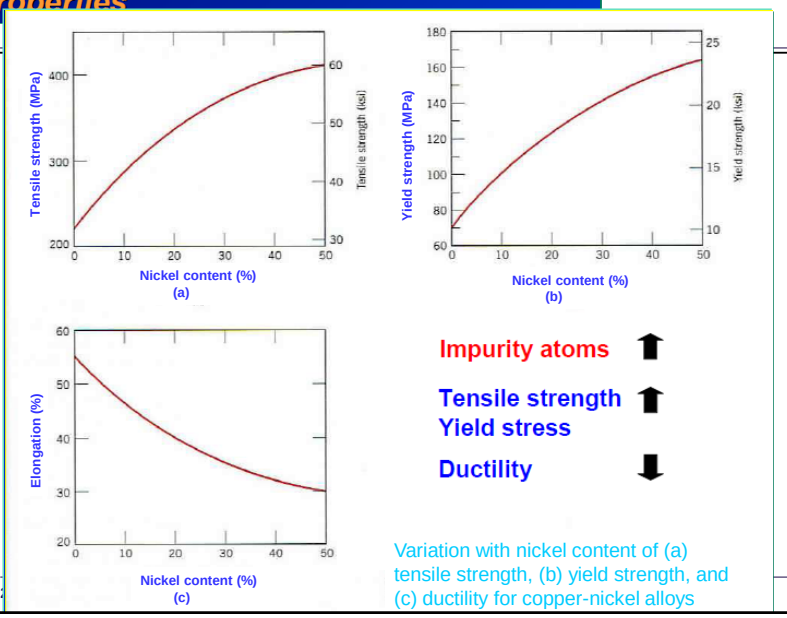
Solid Solution Strengthening

- Impurity that go into solid solution impose lattice strains on surrounding host atoms.
- Lattice strain field interactions between solid solution and impurity atoms result in restriction of dislocation movement.
- This is one of the most powerful reasons to make alloys, which have higher strength than pure metals.



Example:
24k gold is too soft. If we put in 16% silver and 9% copper, we get an alloy that looks just like pure gold, but is much more strong and durable. We call this 18k gold. (18/24 = 75% gold)

Effects of solute alloy additions on tensile properties



Impurity atoms ⬆

Tensile strength ⬆
Yield stress

Ductility ⬇

Variation with nickel content of (a) tensile strength, (b) yield strength, and (c) ductility for copper-nickel alloys

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3.5 Strengthening Mechanism

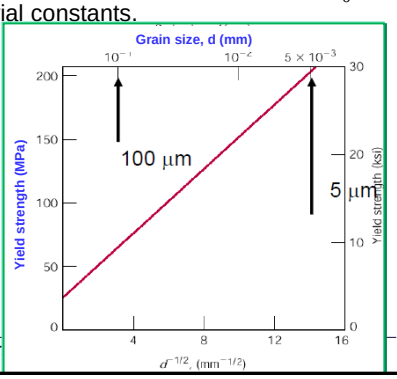
Fine particle strengthening

- Making a smaller grain size increases strength (more obstacles and shorter slip distance.)
- This can be quantified by the so-called "Hall-Petch Equation:

$$\sigma_y = \sigma_0 + K_y d^{-1/2}$$

Where σ_y is yield strength, d is grain size, σ_0 and K_y are material constants.

Strength triples as grain size goes from 100 μm to 5 μm.



- "nanostructured metals", which have grain sizes from 20 to 200 nm. They can have very high strength.

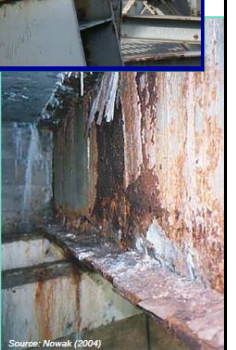
Chapter Three: Metal

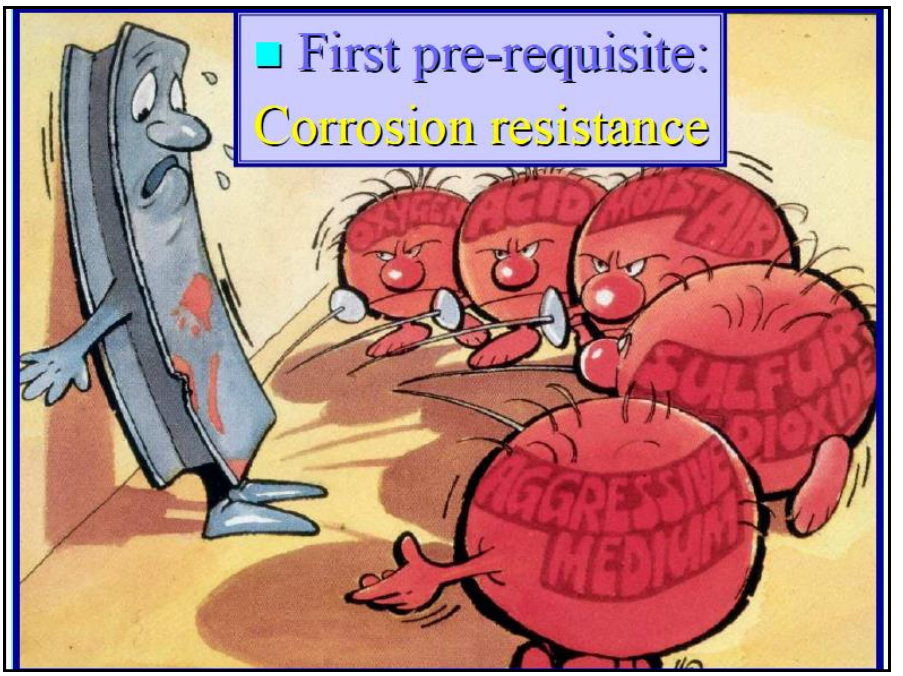
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3.6 Corrosion

- Corrosion is a destruction of a material by electrochemical reaction. When the steel corrodes, rust is formed.
- Some rust on the steel reinforcement before placing of concrete is OK
- Rust on the steel structures must be avoided as it can lead to reduction in strength.
- Steel rust at the rate of about 0.5mm/year.







3.6 Corrosion

- Corrosion – degradation of material properties caused by surrounding environment.
- Corrosive environment – water solution of oxygen and salts, soil, atmosphere…
- The amount of time the steel stays wet affects the rate of corrosion.
- Environmental contaminants may accelerate corrosion. Examples are SO_2 in acid rain, and salts (from sea or deicing salts).



3.6 Corrosion

In order for rust to occur, we need 4 elements:
Anode: The electrode where corrosion occurs.
Cathode: The other electrode needed to form a corrosion cell.
Conductor: A metallic pathway for electrons to flow.
Electrolyte: A liquid that can support the flow of Electrons.

Steel by itself already has 3 elements, it only needs water (electrolyte) to complete the corrosion cell.

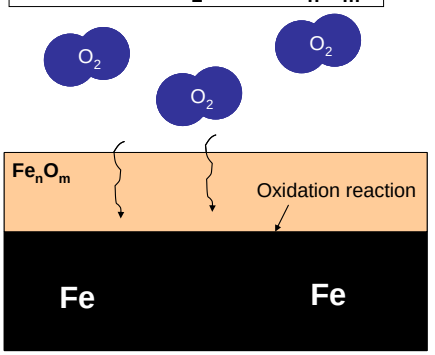
Reactions:

Anode Side	Fe	→	$Fe^{2+} + 2e^-$
	$Fe^{2+} + 2(OH)^-$	→	$Fe(OH)_2$
			Ferrous Hydroxide (Black Rust)
	$4Fe(OH)_2 + 2H_2O + O_2$	→	$4Fe(OH)_3$
			Ferric Hydroxide (Red Rust)
Cathode Side	$4e^- + 2H_2O + O_2$	→	$4(OH)^-$

3.6 Corrosion

Mechanism: Dry Corrosion

$$2nFe + mO_2 \rightarrow 2Fe_nO_m$$



- Dry corrosion takes place at the oxygen-metallic interface;
- Aluminum, chromium, and nickel are metals whose oxides tend to form a good protective layer that inhibits further corrosion;
- Generally, iron oxides have a much lower mass density than that of iron. Hence, they do not adhere well to steel/iron.

3.6 Corrosion

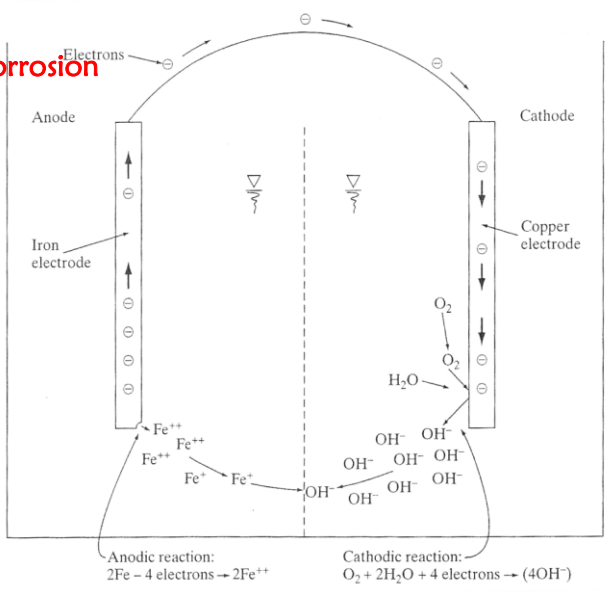
Passivation

Formation of layer of corrosion product (oxide, hydroxide), which protects the rest of material against the corrosion Al, Ti: are covered by compact layer of oxides (Al_2O3, TiO_2)
Passive layer: 1 – 10 nm

3.6 Corrosion

Mechanism: Wet Corrosion

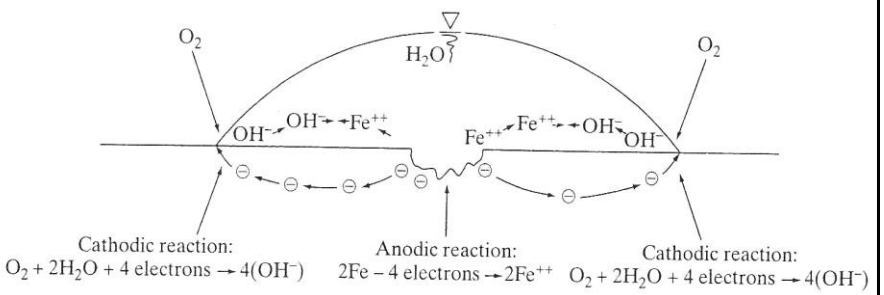
- Each metal has what is called an electrode potential. A metal that has a high electrode potential tends to function as a cathode, and a metal with a lower potential will function as an anode and thus corrode.



3.6 Corrosion

Mechanism: Wet Corrosion

Corrosion on metal surface exposed to a humid environment



3.6 Corrosion

Mechanism: Wet Corrosion

- Driving force of corrosion – difference of electrode potentials of anode and cathode

- electrode potential – depends on type of metal (material property) and composition of electrolyte (concentration of O_2, pH)

- electrode potential series of metals – more noble metals are more stable, less noble metals are corroded

\- Mg Al Zn Cr Fe Co Ni Sn Pb H Cu Ag Au +
less noble (corrode) more noble

3.6 Cor

More likely to corrode

less likely to corrode

Metal	Reaction	Electrode Potential (V)
Li	$Li \leftrightarrow Li^{+}+e^{-}$	-2.96
K	$K \leftrightarrow K^{+}+e^{-}$	-2.92
Na	$Na \leftrightarrow Na^{+}+e^{-}$	-2.71
Mg	$Mg \leftrightarrow Mg^{++}+2e^{-}$	-2.40
Al	$Al \leftrightarrow Al^{+++}+3e^{-}$	-1.70
Zn	$Zn \leftrightarrow Zn^{++}+2e^{-}$	-0.76
Cr	$Cr \leftrightarrow Cr^{3+}+3e^{-}$	-0.74
Fe	$Fe \leftrightarrow Fe^{++}+2e^{-}$	-0.44
Cd	$Cd \leftrightarrow Cd^{++}+2e^{-}$	-0.40
Ni	$Ni \leftrightarrow Ni^{++}+2e^{-}$	-0.22
Sn	$Sn \leftrightarrow Sn^{++}+2e^{-}$	-0.13
Pb	$Pb \leftrightarrow Pb^{++}+2e^{-}$	-0.12
H_2		0.00
Cu	$Cu \leftrightarrow Cu^{++}+2e^{-}$	+0.34
Ag	$Ag \leftrightarrow Ag^{+}+e^{-}$	+0.80
Hg	$Hg \leftrightarrow Hg^{++}+2e^{-}$	+0.86
Cl	$2Cl^{-} \leftrightarrow Cl_2+2e^{-}$	+1.36
Au	$Au \leftrightarrow Au^{+}+e^{-}$	+1.50

Standard Electrode Potentials of Different Elements

3.6 Corrosion

Example: If an aluminum pipe with steel fittings were placed underground in a wet region, what do you expect might happen?

Answer: Since iron has an electrode potential of (-.44V) and aluminum a potential of (-1.70V) we would expect the aluminum pipe to corrode onto the steel fittings.

Example: If zinc and iron were placed underground in a wet environment what would be expected?

Answer: Since zinc has a lower potential (-0.76V) than iron (-0.44V), the zinc would be expected to corrode onto the iron.

3.6 Corrosion

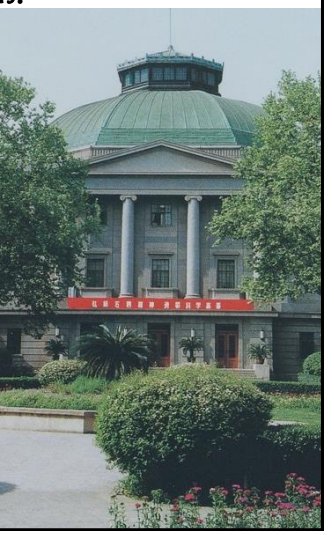
Corrosion of steel and other common metals:

Carbon steel: low corrosion resistance in wet air. In alkaline environment (pH > 9.5), is passivated by iron hydroxides → it enables use of steel as reinforcement in concrete (highly alkaline environment due to $Ca(OH)_2$)

Stainless steel: contains Cr – passivated by Cr_2O_3

Zinc: passivated by $Zn(OH)_2.ZnCO_3$

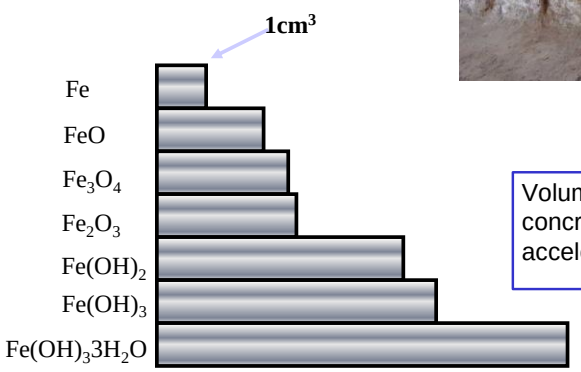
Copper: forms Cu_2O (red), CuO (black) and passive $Cu(OH)_2.CuCO_3$ (green)



3.6 Corrosion



Volume change during corrosion



Volume expansion causes concrete cracking which accelerates corrosion。

3.6 Corrosion

Methods for corrosion resistance

- Modify the environment
- Modify the properties of a metal
- Install a protective coating over the metals
- Impose an electric current to supply electrons
- Change to non-metallic materials

Protective coatings → Paintings
→ Noble metal coating (Cu, Zn, Cr, Ni)
→ Inorganic coating (enamels)

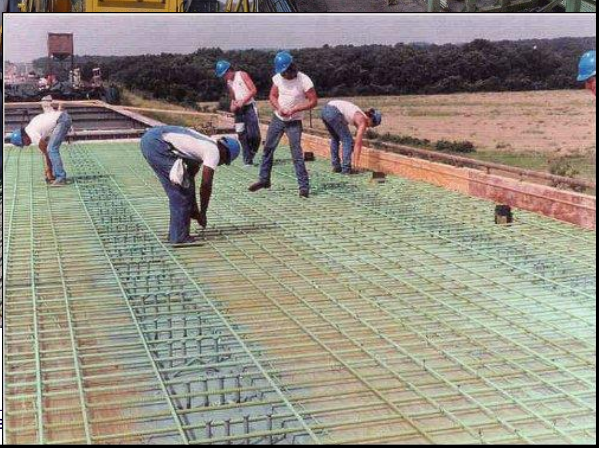
Corrosion inhibitors in the corrosive medium

Electochemical protection → Cathodic
→ Anodic

3.6 Corrosion







3.6 Corrosion

Corrosion Protection

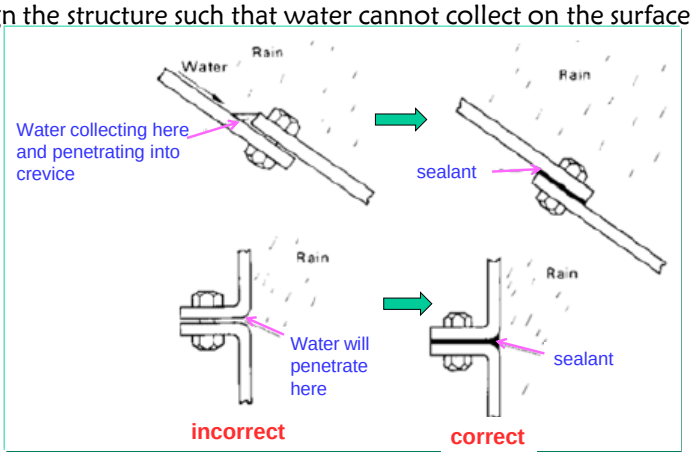


- Chrysler Building built in NYC in 1930 is clad with stainless steel. This helped achieve the art-deco style.
- Stainless steel is a family of steels that contain a minimum of 10.5% chromium.
- Chromium in contact with oxygen forms a chromium oxide film. That is how stainless steel has substantially reduced corrosion.
- So, can stainless steel corrode? Yes! However at a substantially reduced rate when compared to unprotected carbon steel.

3.6 Corrosion

Corrosion Protection

- Design the structure such that water cannot collect on the surface or joints.



3.6 Corrosion

Corrosion Protection

- Design the structure such that inspection and maintenance can be done Easily.







- Applying protective coating to seal off the surface from moisture.
  - The surface to be painted must be dry and clean
  - Periodic repainting is necessary

3.6 Corrosion

Corrosion Protection

- Cathodic protection: we prevent the corrosion of steel by making it the cathode side of the corrosion cell!
  - Sacrificial Anode: this is done by connecting anodic metal with steel. The anode metal will corrode instead of the steel. This anode metal must be replaced occasionally.



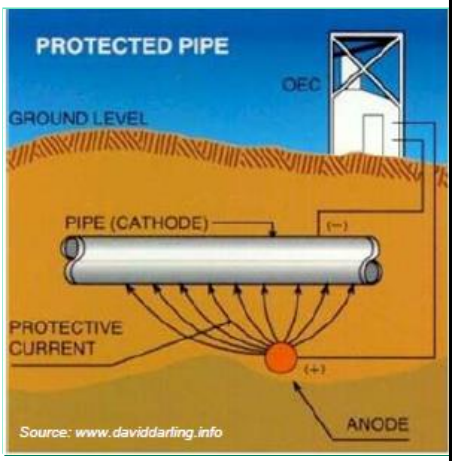
Aluminum anodes are mounted on steel structure

3.6 Corrosion

Corrosion Protection

- Impressed Current Cathodic Protection (ICCP):

Using external power source to make the anode. Inert Anodes such as carbon, titanium, lead, or platinum are used. This is typically used for large structures, such as buried pipelines, as placing sacrificial anodes at regular intervals is impossible.



3.6 Corrosion > Corrosion protection

Metallic coatings

Thin layer of resistant metal (Zn, Cr, Cu, Ni) covers and protects the carbon steel construction against corrosive environment.

Metal plating : electrolytic deposition of the noble metal.
zinc dipping : the protected construction is dipped into molten zinc.

Chapter Three: Metal

1. Introduction
2. Crystalline structure
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7. **Engineering application**

3.7 Engineering application

Steel products for construction

Reinforcing steel
- Round bar
- Deformed bar
- Prestressing strands

Structural steel
- Hot Rolled steel
- Cold-formed steel
- Built-up Members (Steel plate)

3.7 Engineering application

Reinforcing steel

- Reinforcing bars are obtained by hot rolling of steel
- Two main types:
  - **Round bar** – the surface of the bar is smooth
  - **Deformed bar** – the surface of the bar has ribs on it









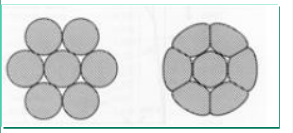
3.7 Engineering application

Reinforcing steel ---- Prestressing strands

- Prestressing strands are made by twisting 2, 3, 7, or 19 wires together. The most common type is the 7-wire strand.
- They are made of higher strength steel than those used in regular reinforcing steel bars.

Type	Size (Diameter)		Shape
	mm	in.	
Two-wire strand	2.9 x 2	0.114 x 2	
Seven-wire strand	6.2 ~ 15.2	0.250 ~ 0.600	
Nineteen-wire strand	17.8 ~ 21.8	0.700 ~ 0.860	

7 wire prestressing strand



Standard strand compacted strand

3.7 Engineering application

Reinforcing steel ---- Prestressing strands



Prestressed concrete girders



3.7 Engineering application

Steel products for construction

Reinforcing steel
- Round bar
- Deformed bar
- Prestressing strands

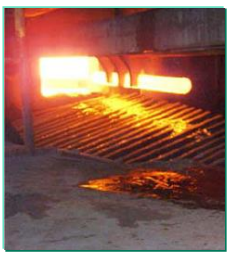
Structural steel
- Hot Rolled steel
- Cold-formed steel
- Built-up Members (Steel plate)

3.7 Engineering application

Structural steel ---- Hot Rolled

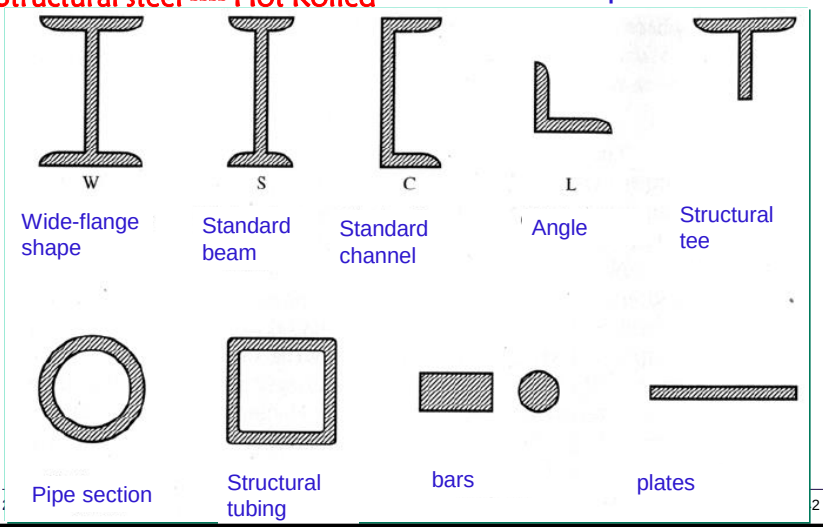
- Hot rolled shapes are obtained by passing very hot block of steel through various rollers several times to obtain the desired shape.
- Typical shapes are W or H (wide-flange), I, C (Channel), L (Angle), T, Pipe, and Tube.





3.7 Engineering application

Structural steel ---- Hot Rolled **Standard rolled shapes**



3.7 Engineering application

Structural steel ---- Hot Rolled

They are used for main structural members, such as truss members, beams, and columns.

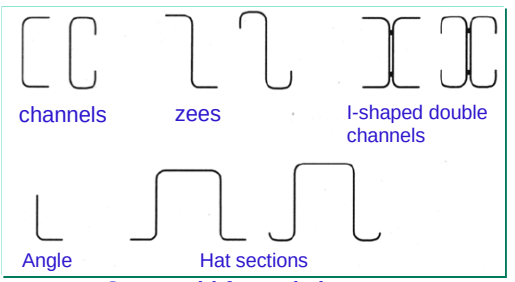




3.7 Engineering application

Structural steel ---- Cold formed

- Cold-formed shapes are obtained by stamping and/or bending steel plate to a desired shape at normal temperature.
- Cold-formed sections usually have small thickness (we cannot bend a very thick plate). Common shapes are C (Channels), Z (Zees), and L (Angles)



Some cold-formed shapes

3.7 Engineering application

Structural steel ---- Cold formed

- They are used mostly for nonstructural elements or for structures that carry small loads

Yield Strength Minimum (MPa)	Tensile Strength Minimum (MPa)	Elongation, Minimum (%)	
		Thickness < 5 mm	Thickness > 5 mm
245	400-510	21	17



3.7 Engineering application

Structural steel ---- Built up members

- We can create structural members by welding steel plates into any desirable shapes
- Typical examples are bridge girders and columns of tall buildings





Summary

1. Strengthening Mechanism

Why?

Cold working, Solid solution, Fine particle strengthening

2. Corrosion

Four elements for corrosion to occur?

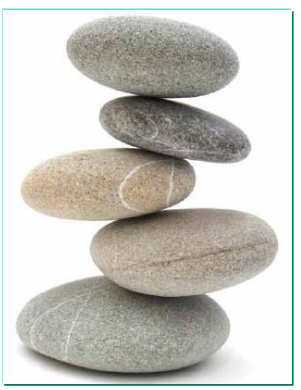
How the corrosion happens?

Protection of corrosion?

3. Engineering Application

Reinforcing steel

Structural steel



QUESTIONS?